

Final Project Proposal - Study Buddy Pupper

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Motivation

It is 11 p.m., midterms are tomorrow, and a student has been sitting at their desk for three hours, but somehow their phone has ended up in their hand seventeen times. That moment captures two problems happening during the long solo study session: people reach for their phones almost automatically, often before they even realize it, and studying alone can feel isolating. However, current focus apps like Forest, Opal, and built-in Focus Modes mainly work after the user has already made the decision to get distracted, and they do not really provide any emotional connection or sense of companionship.

Because of this, we propose Study Buddy Pupper, a small robotic study companion whose visible wellbeing depends on whether the user stays focused. The idea is that one embodied interaction can help address both distraction and loneliness at the same time.

Related Work

A physical robot creates a very different experience from a character on a screen, even if both are technically doing the same task. Research in HRI has shown that robots physically present in the room tend to create stronger engagement and attention compared to virtual or video-based agents [1, 2]. Part of this comes from anthropomorphism: people naturally assign emotions and intentions to objects that move in lifelike ways, and able to anthropomorphize them even the robots is not that realistic. For example, something simple like a Roomba often ends up being named and cared about by its owner after living in the home for a while [3, 4].

We intentionally chose Pupper to take the form of a dog because people naturally want to care for pets. A dog-like design also allows us to use body posture, movement, and small behaviors in ways that feel familiar and emotionally understandable, without entering the uncanny feeling that humanoid robots sometimes create [5].

Connecting the robot's wellbeing to the user's focus also changes the interaction itself. Instead of relying only on self-discipline, the user feels responsible for taking care of a companion. Kidd and Breazeal's Autom study showed that embodied robots could encourage long-term behavior change more effectively than screen-only systems [6]. Similar patterns appear in studies involving companion robots for children and older adults, where repeated caretaking over time strengthens emotional attachment and engagement [7]. Even simple virtual pets like Tamagotchi were enough to make people consistently pay attention to maintaining the creature's wellbeing [8]. In our case, when Pupper slows down or stops "eating," the user reacts to it as if it were a small companion that needs care.

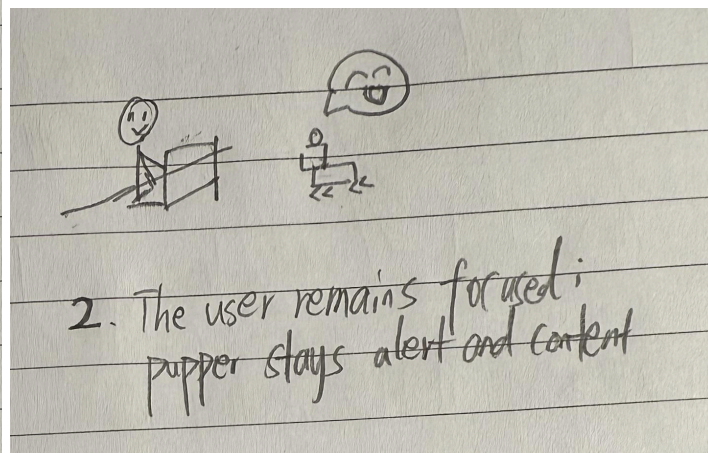
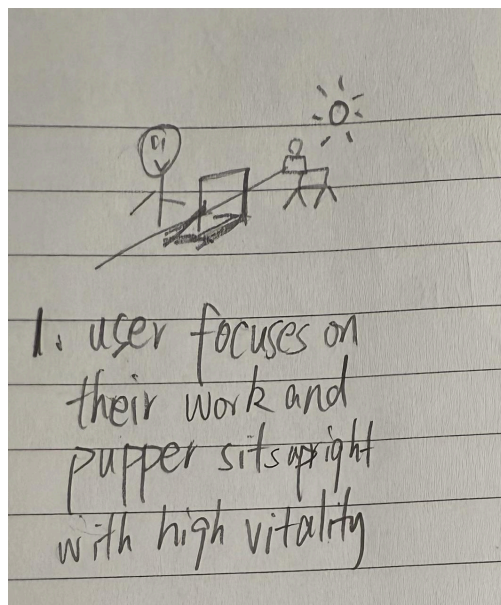
Key Research Questions

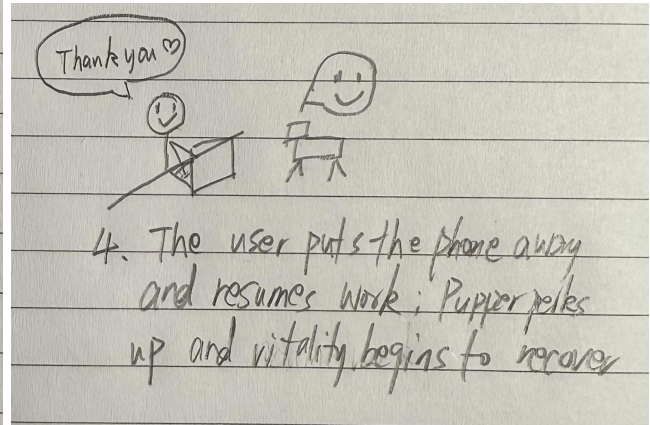
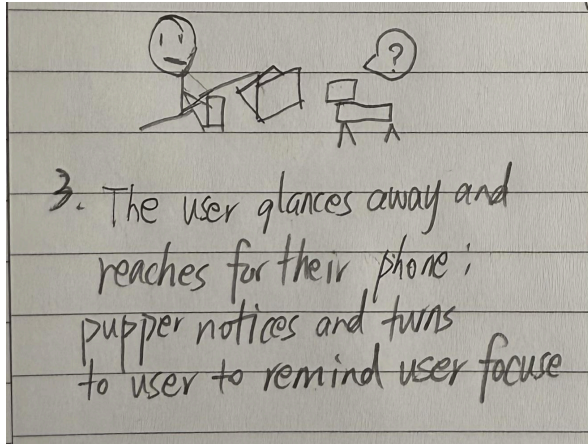
Our central research question is behavioral: does an embodied companion whose wellbeing depends on the user's focus reduce off-task behavior more than self-discipline alone? We frame this question in contrast to apps, which only act on conscious decisions and cannot intervene at the unconscious moment when the user first reaches for their phone.

Main Goals and Interaction Design

Our main goal is to build a Pupper prototype that closes a sense–express–intervene loop in real time during a study session. Pupper watches the user through an onboard camera, expresses how it feels through body posture, face (screen), and short vocalizations, and steps in with proximal behaviors such as walking toward the user or whimpering when the user has been distracted for too long.

Overview Figures





Project Plan

The Study Buddy is a physical, embodied virtual pet whose wellbeing depends on the user's focus. Our goal is to leverage the gamification capabilities of robots to guide users and help them stay focused on the task at hand, with the robot taking on the role of a supportive companion and gentle leader. The context of use is a personal study or work desk, where a single user engages in focused individual work over Pomodoro-style sessions. Using a Pupper robot as our platform, the system combines computer vision and simple speech recognition with timer-based session structuring to create an interactive accountability partner that responds emotionally to user behavior.

The interaction unfolds across a full study session. To onboard, the user names their Study Buddy, which begins with a vitality score of 50/100. The vitality score is session independent and determines the robot's mood and behavior tier. To begin a session, the user sets the timer by pressing buttons on the robot and places their phone in a location visible to the robot's camera. Optionally, they can also input to-do list to be displayed in the robot. The user then studies normally at their desk while Pupper monitors their behavior. The necessary inputs include camera feeds for detecting user presence at the desk, gaze direction, and phone position, along with button inputs for timer configuration and voice input for managing a to-do list during the session. The user can also say simple commands such as "pause session" to the robot during the session.

In response to these inputs, Pupper exhibits a range of behaviors. When the user becomes visibly distracted, for instance, by picking up their phone or leaving the study zone, the robot displays sad puppy eyes and asks, "Are you sure you'd like to interrupt the zone session?" in an attempt to coax the user back into focus. If the user leaves the desk, Pupper grants a 5-minute grace period; failing to return within that window pauses the session and leaves the pet in a dejected state, which reduces its vitality to a suboptimal state. When the preset timer completes successfully, Pupper gains vitality points. Break periods are when Pupper thrives: it dances and adds vitality

point alongside the user, reframing breaks as positive bonding time rather than mere rest. Beyond these reactive behaviors, Pupper can look at the user, look at the work surface, turn its head to track user movement, display happy, sad, and worried expressions through different head positions, and perform greeting and sleep behaviors at the start and end of each session.

The role of the robot is that of an emotionally responsive study companion whose wellbeing is tied to the user's discipline, creating a sense of mutual care. Information is conveyed to the user through the robot's physical expressions (head position, movement, dance), vocal prompts, and vitality state. We plan to evaluate the system in a human-focused way by running study sessions with participants who had previously used app oriented pomodoro timers and measuring subjective measures (perceived helpfulness, emotional attachment to Pupper, willingness to continue using the system).

As stretch goals, we plan to support more nuanced inputs for excusing users mid-session rather than relying solely on the simple 5-minute grace period. We also intend to integrate an LLM for robust speech recognition and customization, which would handle fuzzy voice input and generate time estimates for individual tasks to enable customized study plans.

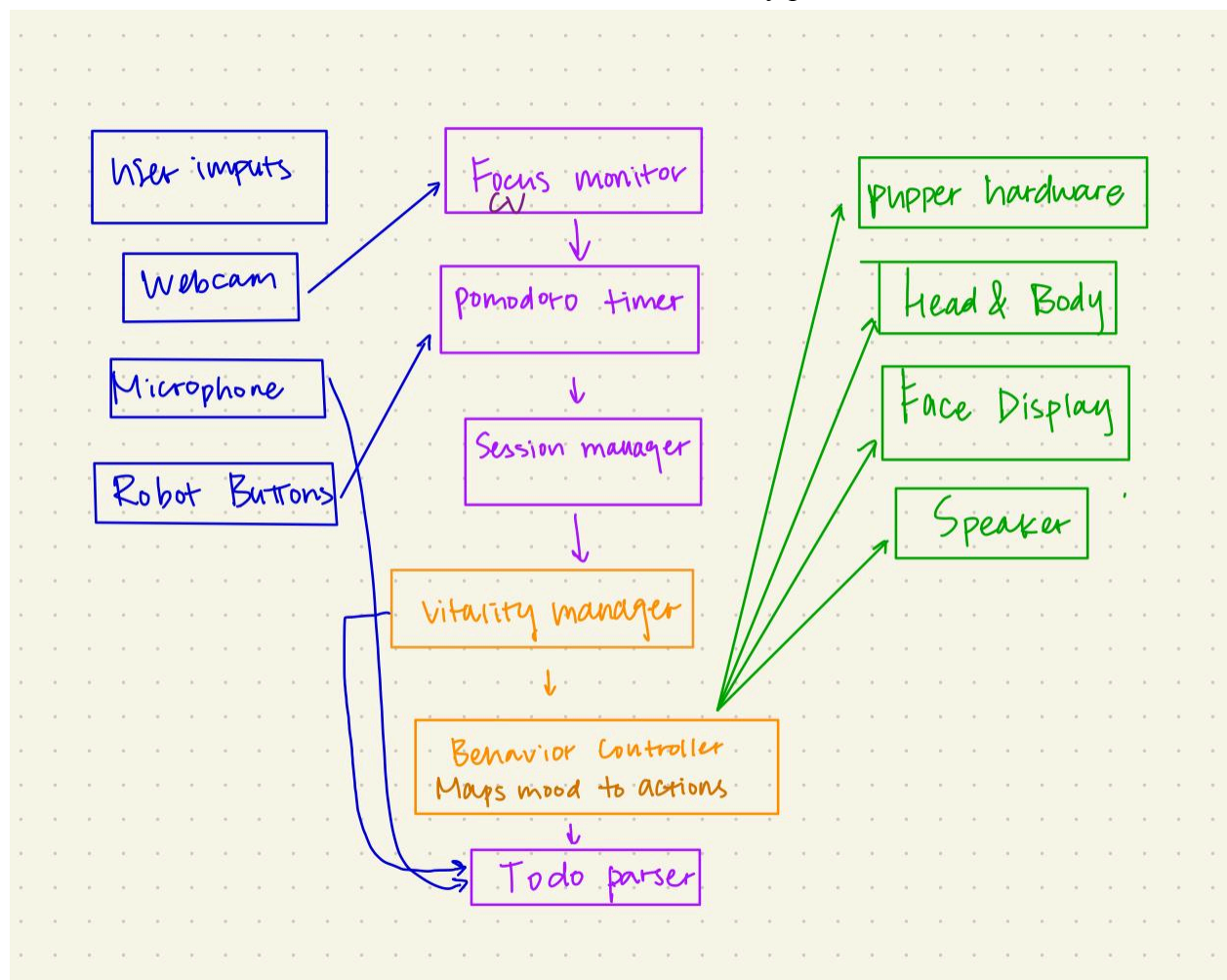


Figure 2: Conceptual diagram showing the main project components

Task Name	Description	Leader
CV research	Set up the pipeline for CV	Juan Yin
CV Phone detection	Train to detect cell phone, test under different lighting, detect if cell phone moved to be picked up	Lele Zhao
CV User Detection	Train to detect user's upper body and arms, detect if user present or not, handle edge cases like being partially in frame or changing positions	Juan Yin
CV Attention Detection	Train to detect user's gaze, detect if eyes looking at their work or away, setup a threshold in case they are looking away for a different reason, test for edge cases like glasses flare	Juan Yin
Game/behavior design/point system + music	Define all behaviors (such as idle, being happy, being sad, dancing), define when to do such behaviors implement point system and when points will be lost or gained Find audio to play for different scenarios, test audio output	Nicole Go
Pupper Motion + pomodoro timer	Define how the pupper will move and when, and if expressions will change Implement pomodoro timer for studying and breaks,	Nicole Go

	implement functions to start timer and not punish the user for exceeding 5 minutes of absence	
Speech recognition	Allow speech detection so user can input todo lists. Allow users to input simple commands such as pause or resume	Lele Zhao
Design Evals	Design form for study session feedback and questions for semi structured interview	Everyone

Figure 3: detailed breakdown of tasks

HRI form

Task Name	Description	Type of Evaluation: Human-Focused or Robot-Focused	Leader
Ethical Approval and Participant Consent	Create consent forms for the participants to understand and sign	Human-Focused	Everyone
Participant Recruitment	Recruit participants to partake in this participant in UCSD	Human-Focused	Everyone
Conduct Robot Interaction Sessions	Design the study area with the pupper, keeping external parts	Human-Focused	Everyone
Post-Experiment Survey	Give surveys to participants	Human-Focused	Juan Yin
Test Audio Quality	Test if the conversation is carried out recognizably	Human-Focused	Nicole Go
Test Display Image Visibility	Test if pupper's display functionality aligns with the design	Human-Focused	Juan Yin

Test Speech Recognition	Test if it can input to-dos Test if it can recognize simple commands	Human-Focused	Lele Zhao
Equipment Setup	Set up phone, hardware and pupper	Robot-Focused	Lele Zhao
Test Behavior	Test if the behavior align with the design	Robot-Focused	Nicole Go
Edge Cases and Error Handling	For all components	Robot-Focused	Everyone

Figure 4: Detailed breakdown of main evaluation tasks

Planned Experimental Methodology

For robot-focused evaluation, we must test equipment set up, testing behavior, and error handling and edge cases. This will be tested through quantitative data and metrics that define how successful and efficient each aspect is [8]. To test equipment setup, we want to ensure all necessary equipment is there and all variables are controlled in the environment where the participant will interact with the robot. We will count how many possible variables need to be controlled and ensure all are. We will count the number of items needed for equipment as well and ensure all are present. For testing behavior, we will create a list of all the robot's functions, and count how many times it performs expected behavior after its triggers. An instance of this is counting the number of times it successfully detects an object. For testing edge cases, we intend on repeatedly putting the robot under specific edge conditions (e.g. different lighting) and counting the number of times it is successful. For all of these, we use success rate.

For human-focused evaluation, we must ensure participant consent, participant recruitment, , conducting Robot interaction sessions, creating a post-experiment survey, audio quality, display image visibility, and speech recognition. To do all of these and assess user experience, we want to define constructs such as how much the robot encourages engagement, trust in the robot, how much they trust the robot, and how much they view the robot as a pet [8].

Due to being restricted by limited participants and individual experiences while studying, we will have a within-subjects design [10]. A prerequisite for people to participate is that they must have used another study tool before, such as a study app. We will provide a demo of all the robot's functions. Participants will have the opportunity to interact with the robot for 15 minutes while completing a study task such as filling out an assignment. The robot will start out in its desk zone and the participant can use it however they please. While they interact with the robot, we will do descriptive research by observing how they interact and if it seems to help the person [9]. We

won't disrupt so that they can perform the task as naturally as possible. We will observe if the person seems to treat the robot as a pet and if they seem to feel encouraged by it, all through qualitative research. Finally, we will give out questionnaires to compare other study methods with our robots. In the databases, there are questionnaires for trust, socialness, anthropomorphism, usability, and attitudes. We will ask them questions for each study method. Specifically, anthropomorphism is important since they would more see the robot as a pet [3].

Gantt Chart

Item	Qty	Purpose	Est. Cost	Source	Status
Pupper robot	1	Embodied platform; locomotion, gait, and posture as expressive channels	\$0 (lab)	lab	Borrow from class
Raspberry Pi 4 / 5 onboard compute	1	Runs CV pipeline, behavior controller, audio I/O	\$0 (lab)	Included with Pupper	Borrow from class
Pi Camera Module v2 (or USB webcam)	1	Visual sensing for user presence, gaze direction, and phone position	\$0 (lab)	lab	Borrow from class
USB microphone (compact, omnidirectional)	1	Captures voice input for to-do list and pause/resume commands	\$15–\$25	Amazon	Need
Small USB or 3.5 mm speaker	1	Plays vocalizations, alert tones, and break-time music	Free	—	Have
Powered USB hub (4-port)	1	Connects mic, speaker, and auxiliary peripherals to the Pi	Free	—	Have
OpenCV + YOLOv8 (phone & user detection)	—	CV pipeline	Free	Open source	Have
Whisper (local) or Web Speech API	—	Speech recognition for to-do list	Free	Open source	Have
LLM API credits (stretch goal)	—	Fuzzy task parsing and time estimation	\$5	OpenAI / Anthropic	Need if pursued

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